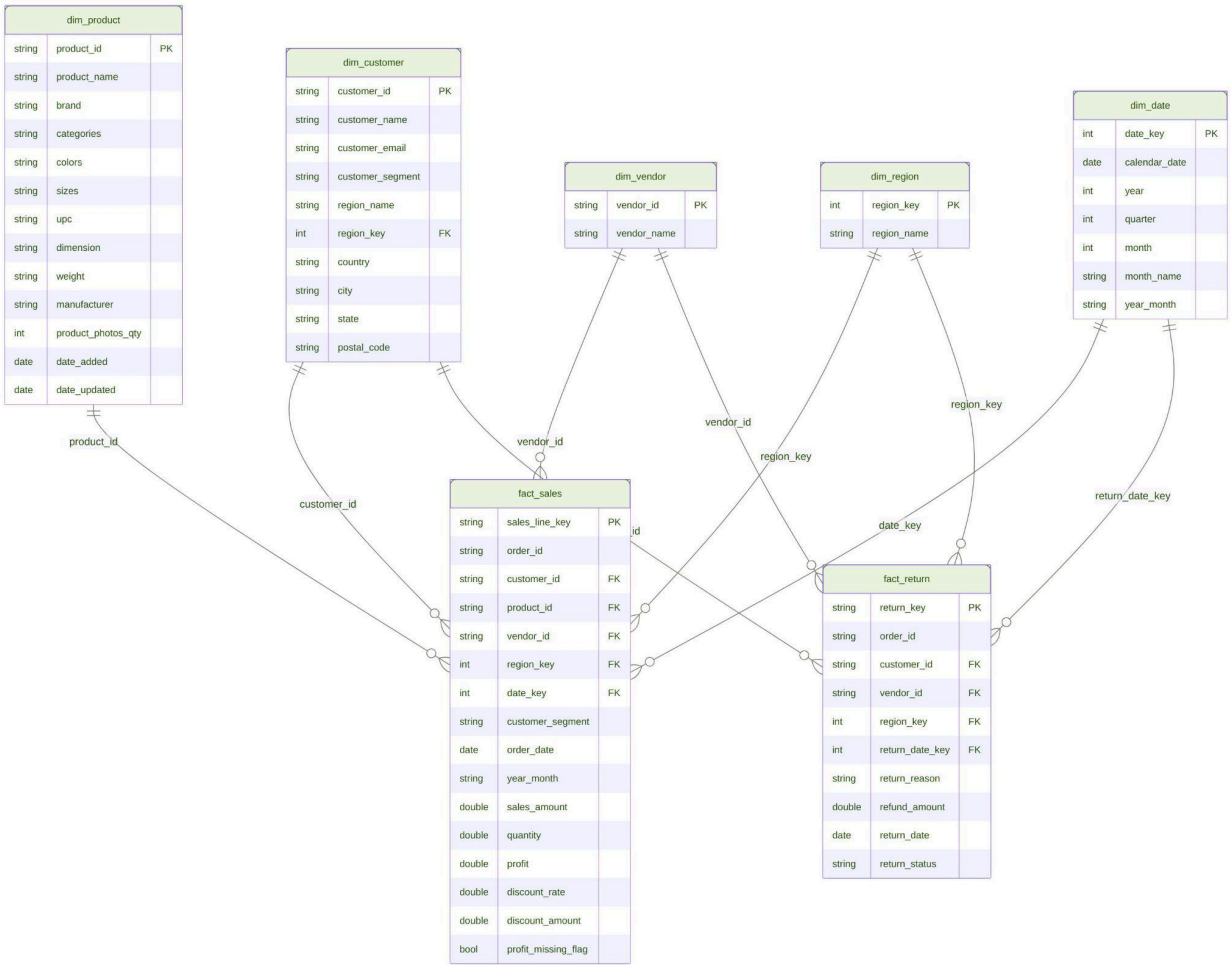
# Dimensional Model

We designed and implemented a complete **Star Schema dimensional model** for the Gold Layer of the analytics solution.
The model enables fast, reliable reporting for sales and return behavior across customers, products, vendors, and regions.

Below is the final dimensional model diagram I produced:




This schema includes:

- **2 Fact Tables:**
  - `fact_sales`
  - `fact_return`
- **5 Dimension Tables:**

    - `dim_customer`
    - `dim_product`
    - `dim_vendor`
    - `dim_region`
    - `dim_date`

# 2. Fact Table Definitions

## 2.1 fact_sales

**Grain:** One row per sales transaction line.
**Measures:**

- sales_amount
- quantity
- profit
- discount_rate
- discount_amount
- profit_missing_flag (data quality metric)

**Foreign Keys:**

- customer_id
- product_id
- vendor_id
- region_key
- date_key

## 2.2 fact_return

**Grain:** One row per return/refund event.
**Measures:**

- refund_amount

**Descriptive Fields (Degenerate Dimensions):**

- return_reason
- return_status

**Foreign Keys:**

- customer_id
- vendor_id
- region_key
- return_date_key

# 3. Dimension Table Definitions

## 3.1 dim_customer

**Primary Key:** customer_id
**Attributes:** name, email, segment, region, city, postal_code, etc.
**Source:** `silver_customer`

## 3.2 dim_product

**Primary Key:** product_id
**Attributes:** product_name, brand, category, size, weight, UPC, manufacturer, etc.
**Source:** `silver_product`

## 3.3 dim_vendor

**Primary Key:** vendor_id
**Attributes:** vendor_name
**Source:** `silver_vendor`

## 3.4 dim_region

**Primary Key:** region_key
**Attributes:** region_name
**Source:** `silver_region`

## 3.5 dim_date

**Primary Key:** date_key
**Attributes:** calendar_date, month, quarter, year, year_month
**Source:** `silver_date`

# 4. How My Model Solves PrimeInsurance's Key Business Failures

We were asked to address three critical failures faced by PrimeInsurance: **regulatory pressure, claims backlog, and revenue leakage**.
Below is how my model directly solves these challenges.

## 4.1 Regulatory Pressure

Regulators require accurate, audit-friendly reports.
My design solves this by:

- Implementing **surrogate keys** to allow historical tracking
- Ensuring consistent time-based analysis via the **dim_date** table
- Centralizing all facts in a **single source of truth** for reporting

This improves compliance and transparency.

## 4.2 Claims / Returns Backlog

PrimeInsurance struggles to track claims/returns progress.

My `fact_return` model allows:

- Monitoring of **return_status** and delays
- Identifying bottlenecks by **region**, **vendor**, or **customer**
- Calculating claim aging using **return_date_key**

This helps the operations team reduce backlogs.

## 4.3 Revenue Leakage

PrimeInsurance loses money due to incorrect profitability, discounts, and untracked refunds.

My model fixes this by:

- Capturing **refund_amount** in `fact_return`
- Tracking missing or suspicious values using **profit_missing_flag**
- Analyzing discount patterns across products, vendors, or regions

The business can now detect issues earlier and prevent financial loss.